Roll No.

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem.-2)

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Subject Code : BP-202T

M.Code : 74968

Date of Examination : 30-05-23

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Briefly write about the following :

a) Draw structures of various isomers are possible for $C_4H_{10}O$. Write their chemical names.

b) Define electromeric effect.

c) Comment upon the stability of carbocations.

d) Comment upon the uses of chlorobutanol.

e) Write the chemical equation when ethyl chloride reacts with KCN. Give the name of product formed.

f) Write down cannizaro reaction.

g) Give chemical test to distinguish between acetaldehyde and acetic acid.

h) Give the structure and uses of tartaric acid.

i) Write any two methods of preparation of alkynes.

j) Write down the structure of any two medicinally important alcohols.

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SECTION-B

2. Write a note on different mechanisms involved in the substitution reactions of alkyl halides. Comment upon their reaction kinetics, order of reactivity and stereochemistry involved.

3. a) Give detailed note on the various methods of preparation of aldehydes and ketones.

   b) Write a note on effect of substituents on the acidity of carboxylic acids.

4. Account for the following :

   a) Halogenation of alkanes

   b) Factors affecting E_1 and E_2 reactions

SECTION-C

5. Explain the stability of alkenes and conjugated dienes.

6. Write down various tests involved in the identification of alcohols.

7. Classify organic compounds. Write down atleast one chemical reaction of each class.

8. Write a short note on Benzoin condensation reaction.

9. Define Basicity. Comment upon the effect of substituent upon basicity of various amines.

10. What is the structure and use of Hexamine and vanillin?

11. Write structure of any five medicinally important examples of carboxylic acids.

12. Give chemical tests to identify the presence of propanoic acid, acetaldehyde and methylpropanoate.

13. Comment upon hybridisation in alkanes.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.